

GROW Analysis: Graduate Goals Reflection

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Goal (G)

My primary graduate objective is to establish a dual career path that integrates high-level industry leadership with academic contributions as a resident professor. I aim to develop a specialized expertise in **Vision-Language Models (VLMs)** for spatiotemporal reasoning within the construction sector. To quantify my progress toward this goal, I have set a target to publish **five journal articles** and **five conference papers** prior to my graduation. My immediate milestone is to submit my current research findings—which serve as a foundational training ground for my doctoral journey—to a major industry journal by the end of this summer. Additionally, I aim to ensure my technical versatility remains sharp for the computer science job market by completing **300 LeetCode problems** and obtaining a **professional cloud certification**. These benchmarks are designed to prepare me for continuous industry engagement and internship opportunities starting in my second year.

Reality (R)

I am currently in the inaugural semester of my Ph.D. in Construction Management Technology, having joined the program in January 2026. My current standing is defined by a non-traditional academic background that includes a Bachelor's in Civil Engineering, an MBA in Finance, and an MS in Computer Information Systems. This interdisciplinary foundation allows me to grasp complex system components and business logic, but I am now tasked with synthesizing these areas into a cohesive research identity.

This semester, I am enrolled in three rigorous technical courses that directly support my research goals. In **Foundations of Natural Language Processing**, I am designing semantic representations—skills I recently applied to build “DebateBot,” a Trump vs. Biden debate simulator utilizing cloud platforms, fully local fine-tuning of the Qwen 2.5 0.5B model, and voice cloning. My **Time Series and Applications** coursework provides the mathematical tools for ARIMA modeling. Finally, my **Computer Vision with Deep Learning** studies are equipping me with rigorous mathematical foundations, which I am actively implementing through assignments using methods like Mean Squared Error (MSE), Normalized Cross-Correlation (NCC), Gradient Descent, and Blob detection.

What is working well is my ability to execute tasks once a problem is defined as solvable. However, the primary factor holding me back is **methodological ambiguity**. Identifying the “right” research problem that is both original and solvable is a significant hurdle. Furthermore, balancing the high-velocity requirements of these three technical courses with the need for self-directed research creates a challenging “unstructured” environment that demands strict time management.

Options (O)

To move forward toward my publication and professional milestones, I have identified three strategic options:

- **Option 1: Deliverable Alignment.** I can align the term projects and papers required for my NLP and Computer Vision courses with my primary research interest in construction management. By using the required individual term paper or team project to prototype my spatiotemporal reasoning framework, I can produce peer-reviewed quality work while fulfilling my course requirements. This reduces the friction between my coursework and my summer publication goal.
- **Option 2: Technical Sprint Planning.** I will establish a structured weekly schedule to reach the **300-problem LeetCode benchmark** by allocating specific hours during the upcoming summer break. Additionally, I can pursue my cloud certification during the post-semester window to ensure my technical profile is “industry-ready” for the co-op opportunities I plan to pursue in my second year.
- **Option 3: Systematic Feedback.** To resolve research ambiguity, I will utilize faculty consultation hours to present a structured analysis matrix of my current literature review. My metric for success is to receive validation on my proposed methodology by mid-semester. This proactive engagement will allow me to move from the “thinking deeply” phase into the “execution” phase with greater confidence in the solvability of my research problem.

Way Forward (W)

I am committing to **Option 1: Deliverable Alignment** to streamline my workload and resolve methodological ambiguity, leveraging my recent technical momentum. My specific action is to synthesize the final projects from my NLP and Computer Vision courses to construct the framework for a comprehensive review paper on Vision-Language Models (VLMs) for spatiotemporal reasoning. Specifically, I will build upon the cloud deployment and local fine-tuning techniques (like those used for the Qwen 2.5 0.5B model and voice cloning in my “DebateBot” simulator) to process unstructured construction text. Simultaneously, my Computer Vision project will elevate the foundational mathematics I am currently implementing—such as MSE, NCC, Gradient Descent, and Blob detection—to evaluate advanced

visual feature extraction from dynamic site feeds. I will then merge these prototypes to formulate a novel multimodal framework.

My strict deadline for this objective is **May 15, 2026**. This perfectly aligns with the end of the Spring 2026 semester, allowing me to leverage structured coursework milestones to iteratively build the manuscript.

The definitive evidence of completion will be twofold: holding the finalized, graded project reports from both courses, and possessing the official submission confirmation receipt for the VLM review paper sent to a high-impact construction technology journal.